PORTFOLIO 2 · DATA ANALYTICS

Revenue & Pricing Analysis

UK online gift retailer · Dec 2010 – Dec 2011 · UCI Online Retail

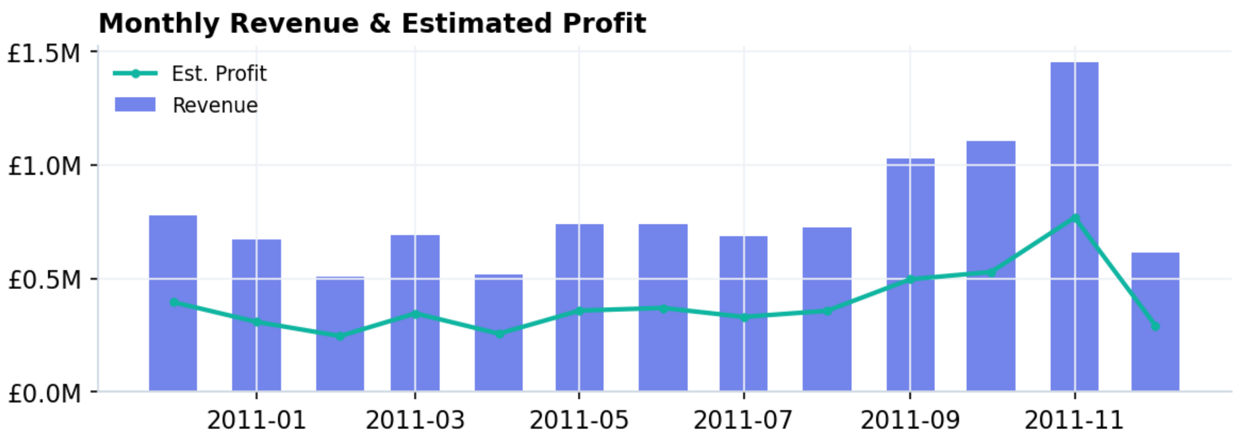
£10.25M	£5.06M	49.4%	£518	£3.28	7.8%
REVENUE	EST. PROFIT	EST. MARGIN	AVG ORDER	AVG PRICE	PROMO SHARE

Project Overview

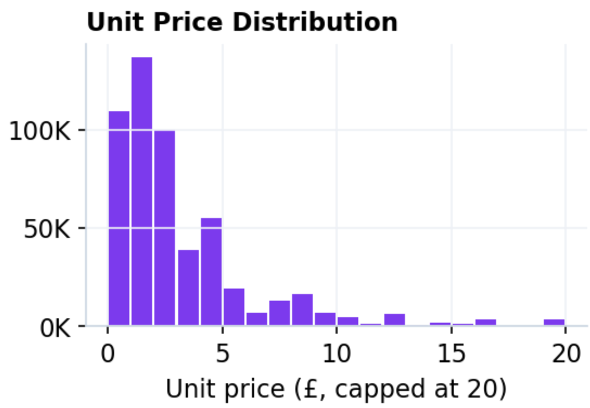
This analysis identifies what drives revenue for a UK online gift retailer and quantifies how pricing and promotions affect volume and margin. It is built on the UCI Online Retail dataset — roughly 542k transaction lines over 12 months.

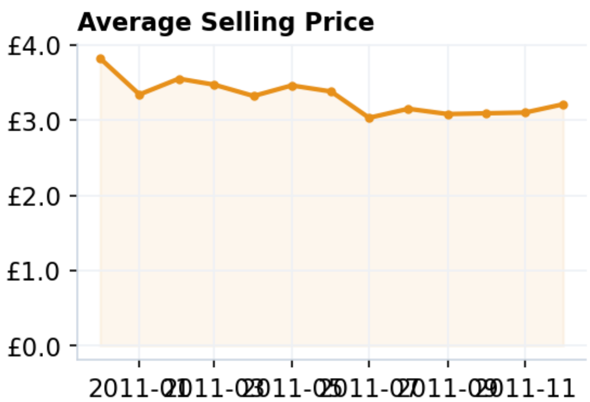
Method. The raw data was cleaned from 541,909 to 522,568 valid line items by removing 5,268 duplicates, 10,587 returns/cancellations, 1,176 invalid prices, and 2,310 non-product service codes. Because the source has no cost or discount fields, discount is derived from each SKU's modal list price, and margin is an estimate assuming cost equals 50% of list price.

Performance at a Glance



Revenue accelerates into the Q4 wholesale season, peaking in November 2011. December is partial (data ends 9 Dec), so its dip is a cut-off artefact.





Key Business Insights

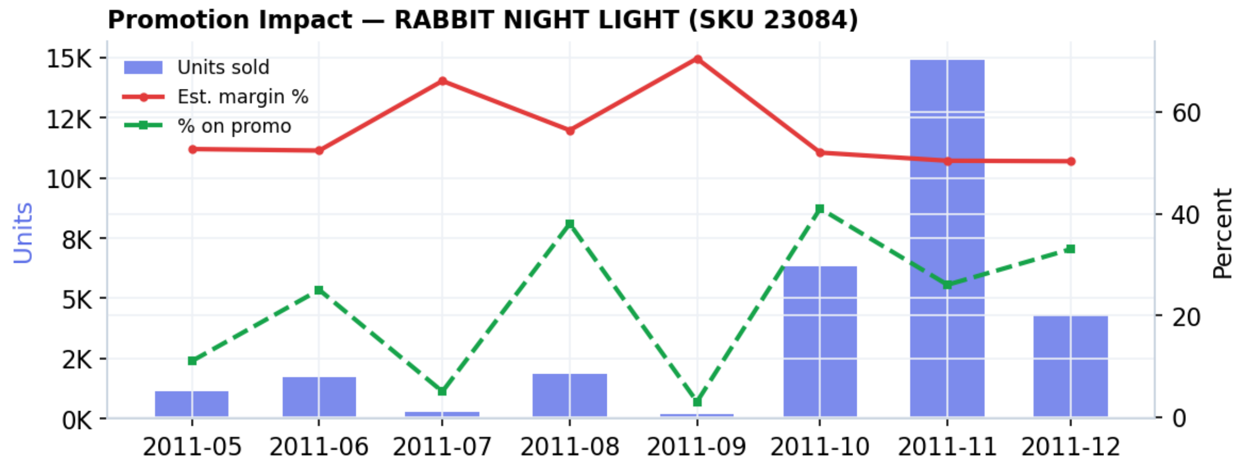
1. **Revenue is volume-driven, not price-driven.**
   At the invoice level, revenue correlates with units sold (0.89) but barely with average price (0.08). The business grows by moving more units, not by charging more.

2. **Pricing is low and tightly clustered.**
   Most items sell for £1-£5 with a long premium tail; average selling price is around £3.28. Small absolute price changes move many units.

3. **Demand is highly seasonal.**
   Revenue climbs sharply from September and peaks in November as wholesale customers stock up for Christmas — a window to plan inventory and pricing.

4. **Promotions lift volume but erode margin.**
   RABBIT NIGHT LIGHT (SKU 23084) sells dramatically more units in promoted months, but estimated margin falls by roughly 15.4 points. Volume is bought with margin.

5. **Revenue is concentrated in a few SKUs.**
   A small set of products drives a large share of revenue, so product-level pricing decisions have outsized P&L; impact.

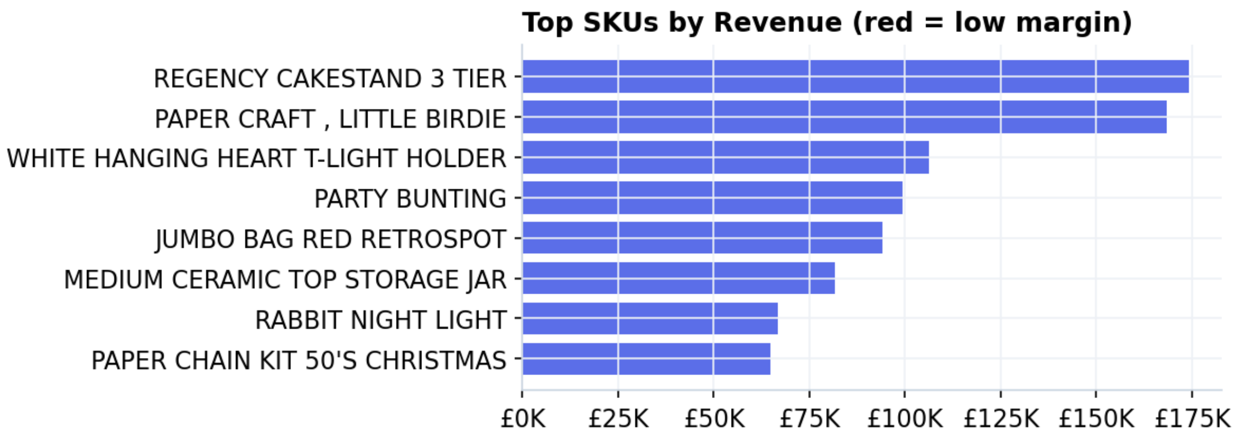


Promotion impact on a real SKU: units (bars) spike when promo share (green) rises, while estimated margin (red) falls.

Recommendations

ACTION	DETAIL	LEVER
Set discounts by product margin	Replace blanket promotions with margin-aware discount ceilings. High-margin SKUs can absorb deeper cuts; thin-margin SKUs should be capped.	Margin

Model promo profitability before launch	For each promotion, require the projected volume lift to more than offset the margin given away — not just a revenue target.	Profit
Plan pricing around Q4 seasonality	Pre-stock top SKUs for the Sep-Nov peak and protect margin by limiting deep discounts during high-demand months.	Ops
Focus pricing attention on top SKUs	Concentrated revenue means a handful of products move the P&L; review their list price and discount policy first.	Growth
Grow via volume levers, not price hikes	Since revenue tracks units, invest in bundling, cross-sell, and availability rather than across-the-board price increases.	Revenue



Deliverables: cleaned dataset (CSV) · Jupyter notebook (.ipynb) · SQL query pack (PostgreSQL) · interactive HTML dashboard · this executive summary · Excel summary. Margin figures are transparent estimates (assumed 50% list-price cost), not source data.